

s -Domain Circuit Analysis

Operate directly in the s -domain with capacitors, inductors and resistors

Key feature – linearity – is preserved

Ccts described by ODEs and their ICs

Order equals number of C plus number of L

Element-by-element and source transformation

Nodal or mesh analysis for s -domain cct variables

Solution via Inverse Laplace Transform

Why?

Easier than ODEs

Easier to perform engineering design

Frequency response ideas - filtering

Element Transformations

Voltage source

Time domain

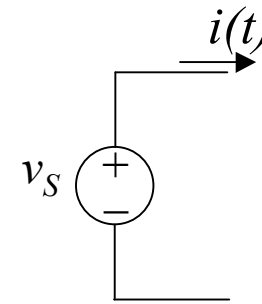
$$v(t) = v_S(t)$$

$$i(t) = \text{depends on cct}$$

Transform domain

$$V(s) = V_S(s) = \mathcal{L}(v_S(t))$$

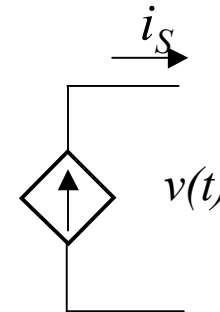
$$I(s) = \mathcal{L}(i(t)) \text{ depends on cct}$$



Current source

$$I(s) = \mathcal{L}(i_S(t))$$

$$V(s) = \mathcal{L}(v(t)) \text{ depends on cct}$$



Element Transformations contd

Controlled sources

$$\begin{aligned}v_1(t) &= \mu v_2(t) &\Leftrightarrow & V_1(s) = \mu V_2(s) \\i_1(t) &= \beta i_2(t) &\Leftrightarrow & I_1(s) = \beta I_2(s) \\v_1(t) &= r i_2(t) &\Leftrightarrow & V_1(s) = r I_2(s) \\i_1(t) &= g v_2(t) &\Leftrightarrow & I_1(s) = g V_2(s)\end{aligned}$$

Short cct, open cct, OpAmp relations

$$\begin{aligned}v_{SC}(t) &= 0 &\Leftrightarrow & V_{SC}(s) = 0 \\i_{OC}(t) &= 0 &\Leftrightarrow & I_{OC}(s) = 0 \\v_N(t) &= v_P(t) &\Leftrightarrow & V_N(s) = V_P(s)\end{aligned}$$

Sources and active devices behave identically

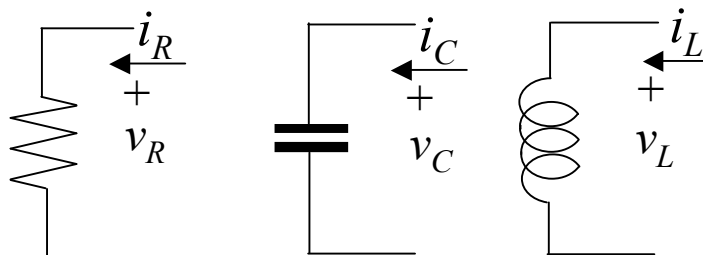
Constraints expressed between transformed variables

This all hinges on uniqueness of Laplace Transforms and linearity

Element Transformations contd

Resistors

$$\begin{aligned} v_R(t) &= Ri_R(t) & i_R(t) &= Gv_R(t) \\ V_R(s) &= RI_R(s) & I_R(s) &= GV_R(s) \end{aligned}$$



$$Z_R(s) = R \quad Y_R(s) = \frac{1}{R}$$

Capacitors

$$i_C(t) = C \frac{dv_C(t)}{dt} \quad v_C(t) = \frac{1}{C} \int_0^t i_C(\tau) d\tau + v_C(0)$$

$$I_C(s) = sCV_C(s) - Cv_C(0^-) \quad V_C(s) = \frac{1}{sC} I_C(s) + \frac{v_C(0)}{s}$$

$$Z_C(s) = \frac{1}{sC} \quad Y_C(s) = sC$$

Inductors

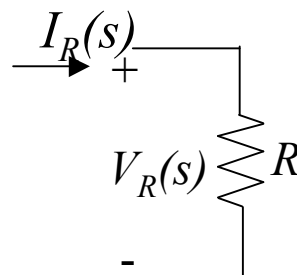
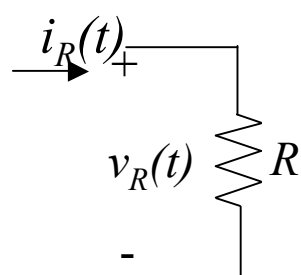
$$v_L(t) = L \frac{di_L(t)}{dt} \quad i_L(t) = \frac{1}{L} \int_0^t v_L(\tau) d\tau + i_L(0)$$

$$V_L(s) = sLI_L(s) - Li_L(0) \quad I_L(s) = \frac{1}{sL} V_L(s) + \frac{i_L(0)}{s}$$

$$Z_L(s) = sL \quad Y_L(s) = \frac{1}{sL}$$

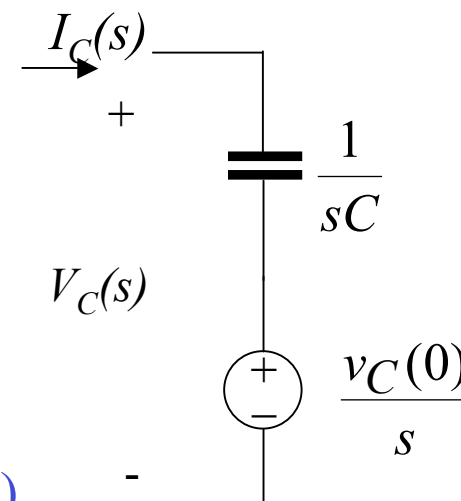
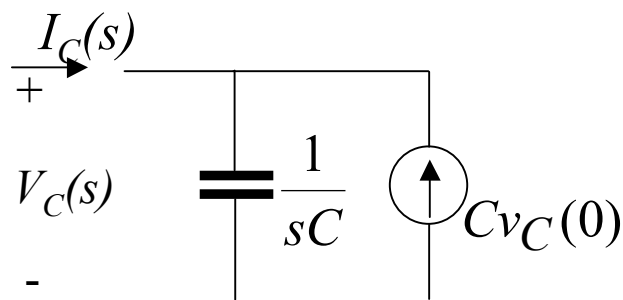
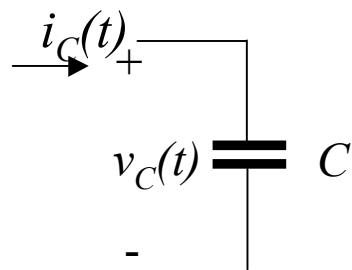
Element Transformations contd

Resistor



$$V_R(s) = RI_R(s)$$

Capacitor



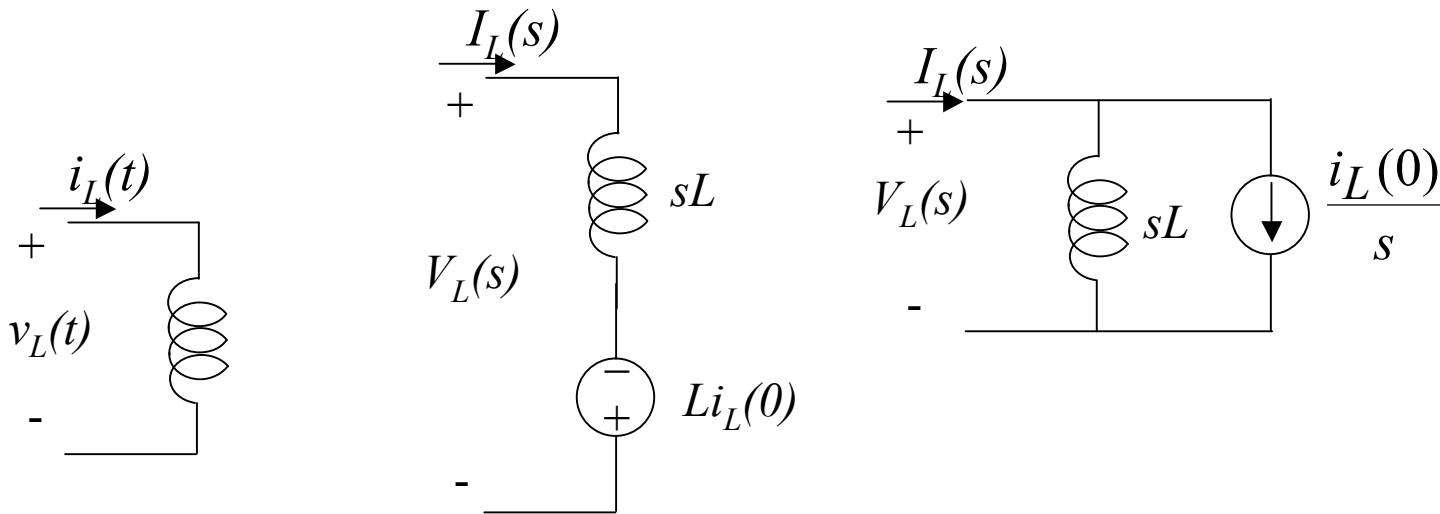
$$I_C(s) = sCV_C(s) - Cv_C(0^-) \quad V_C(s) = \frac{1}{sC}I_C(s) + \frac{v_C(0)}{s}$$

Note the source transformation rules apply!

Element Transformations contd

Inductors

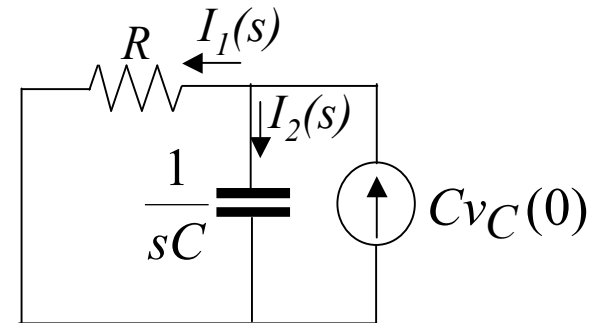
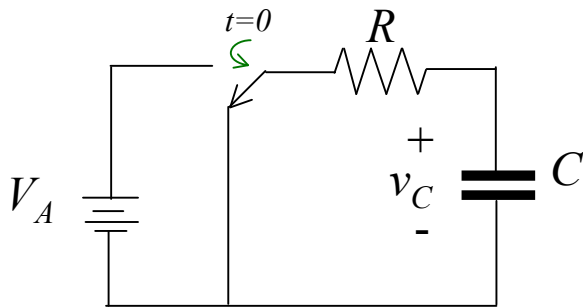
$$V_L(s) = sLI_L(s) - Li_L(0) \quad I_L(s) = \frac{1}{sL}V_L(s) + \frac{i_L(0)}{s}$$



Example 10-1 T&R p 456

RC cct behavior

Switch in place since $t=-\infty$, closed at $t=0$. Solve for $v_C(t)$.



Initial conditions $v_C(0) = V_A$

s-domain solution using nodal analysis

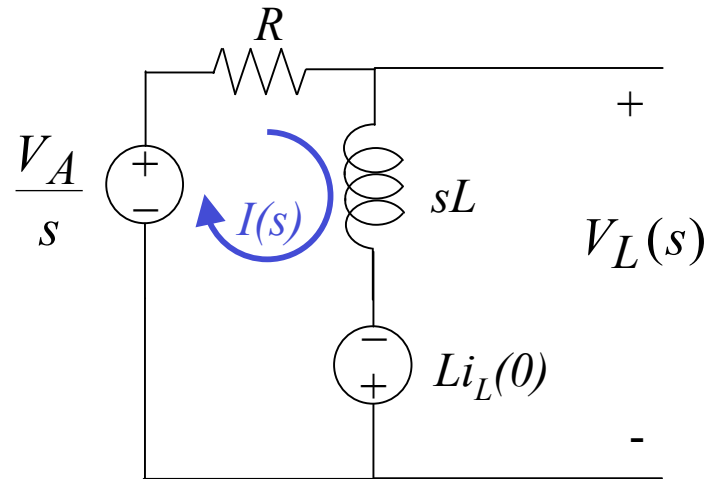
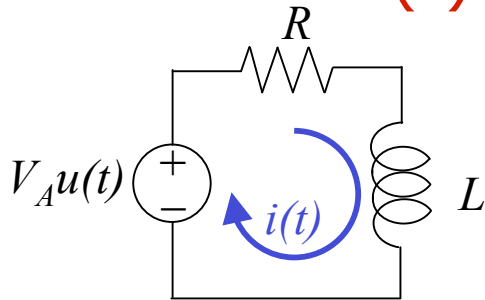
$$I_1(s) = \frac{V_C(s)}{R} \quad I_2(s) = \frac{V_C(s)}{1/sC} = sCV_C(s)$$

t-domain solution via inverse Laplace transform

$$V_C(s) = \frac{V_A}{s + \frac{1}{RC}} \quad v_C(t) = V_A e^{-t/RC} u(t)$$

Example 10-2 T&R p 457

Solve for $i(t)$



KVL around loop $\frac{V_A}{s} - (R + sL)I(s) + Li_L(0) = 0$

Solve
$$I(s) = \frac{V_A/L}{s(s + R/L)} + \frac{i_L(0)}{s + R/L} = \frac{V_A/R}{s} + \frac{(i_L(0) - V_A/R)}{s + R/L}$$

Invert
$$i(t) = \left[\frac{V_A}{R} - \frac{V_A}{R} e^{-Rt/L} + i_L(0) e^{-Rt/L} \right] u(t) \text{ Amps}$$

Impedance and Admittance

Impedance is the s -domain proportionality factor relating the transform of the voltage across a two-terminal element to the transform of the current through the element with all initial conditions zero

Admittance is the s -domain proportionality factor relating the transform of the current through a two-terminal element to the transform of the voltage across the element with initial conditions zero

Impedance is like resistance

Admittance is like conductance

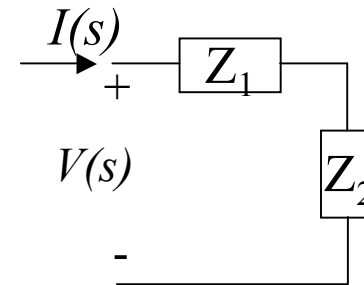
Circuit Analysis in s-Domain

Basic rules

The equivalent impedance $Z_{eq}(s)$ of two impedances $Z_1(s)$ and $Z_2(s)$ in series is $Z_{eq}(s) = Z_1(s) + Z_2(s)$

Same current flows

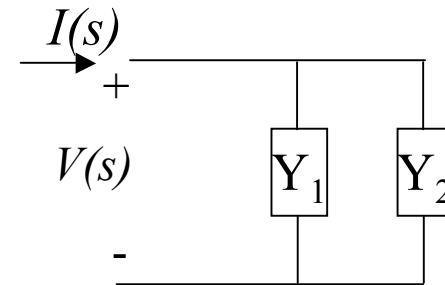
$$V(s) = Z_1(s)I(s) + Z_2(s)I(s) = Z_{eq}(s)I(s)$$



The equivalent admittance $Y_{eq}(s)$ of two admittances $Y_1(s)$ and $Y_2(s)$ in parallel is $Y_{eq}(s) = Y_1(s) + Y_2(s)$

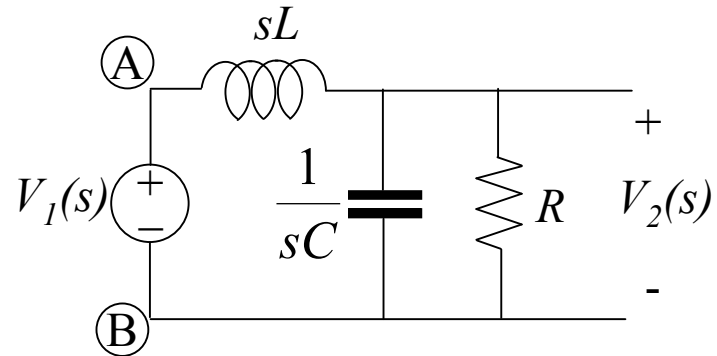
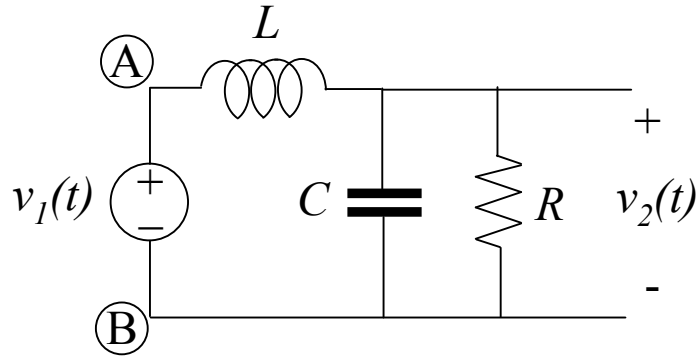
Same voltage

$$I(s) = Y_1(s)V(s) + Y_2(s)V(s) = Y_{eq}(s)V(s)$$



Example 10-3 T&R p 461

Find $Z_{AB}(s)$ and then find $V_2(s)$ by voltage division



$$Z_{eq}(s) = sL + R \parallel \frac{1}{sC} = sL + \frac{1}{\frac{1}{R} + sC} = \frac{RLCs^2 + Ls + R}{RCs + 1} \Omega$$

$$V_2(s) = \left[\frac{Z_1(s)}{Z_{eq}(s)} \right] V_1(s) = \left[\frac{R}{RLCs^2 + sL + R} \right] V_1(s)$$

Superposition in s-domain ccts

The s-domain response of a cct can be found as the sum of two responses

1. The zero-input response caused by initial condition sources with all external inputs turned off
2. The zero-state response caused by the external sources with initial condition sources set to zero

Linearity and superposition

Another subdivision of responses

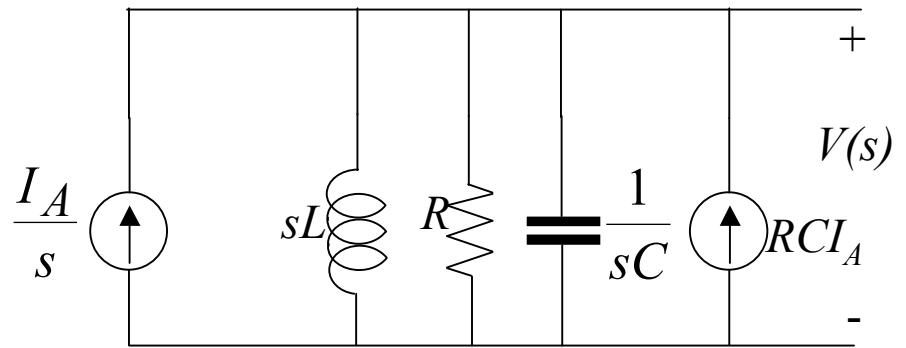
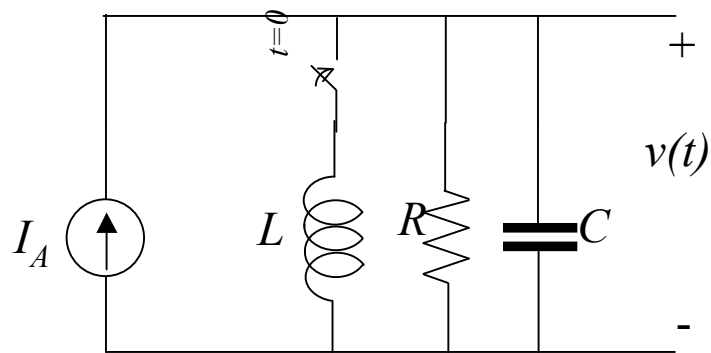
1. Natural response – the general solution
Response representing the natural modes (poles) of the cct
2. Forced response – the particular solution
Response containing modes due to the input

Example 10-6 T&R p 466

The switch has been open for a long time and is closed at $t=0$.

Find the zero-state and zero-input components of $V_2(s)$

Find $v(t)$ for $I_A=1\text{mA}$, $L=2\text{H}$, $R=1.5\text{K}\Omega$, $C=1/6\text{ }\mu\text{F}$



$$Z_{eq}(s) = \frac{1}{\frac{1}{sL} + \frac{1}{R} + sC} = \frac{RLs}{RLCs^2 + Ls + R}$$

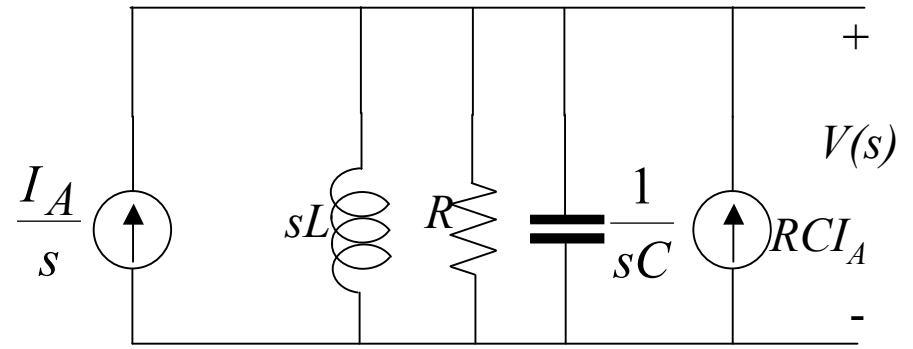
$$V_{zs}(s) = Z_{eq}(s) \frac{I_A}{s} = \frac{I_A/C}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

$$V_{zi}(s) = Z_{eq}(s) RCI_A = \frac{RI_A s}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

Example 10-6 contd

$$V_{zs}(s) = Z_{eq}(s) \frac{I_A}{s} = \frac{I_A / C}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

$$V_{zi}(s) = Z_{eq}(s) RCI_A = \frac{RI_A s}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$



Substitute values

$$V_{zs}(s) = \frac{6000}{(s+1000)(s+3000)} = \frac{3}{s+1000} + \frac{-3}{s+3000}$$

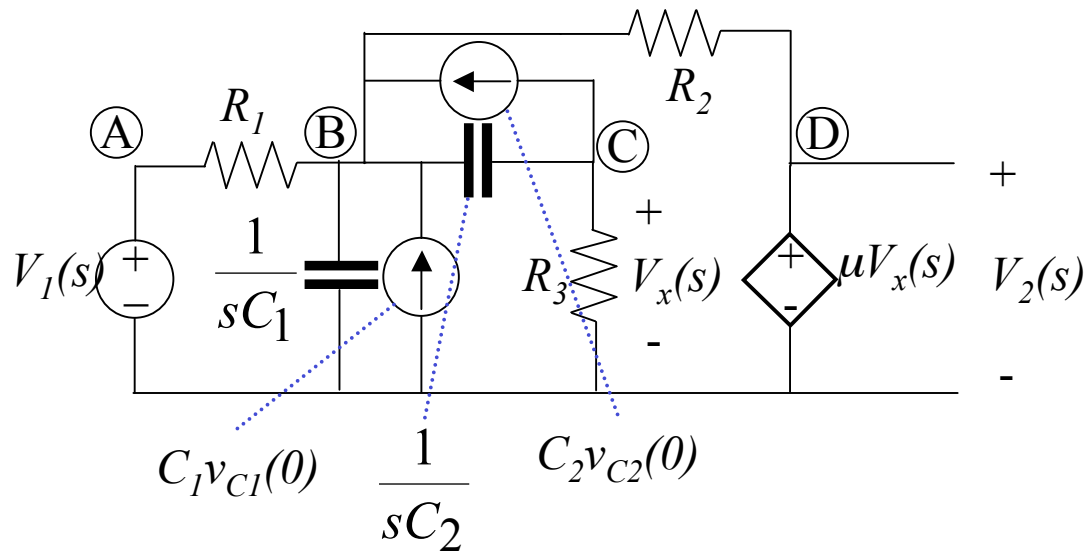
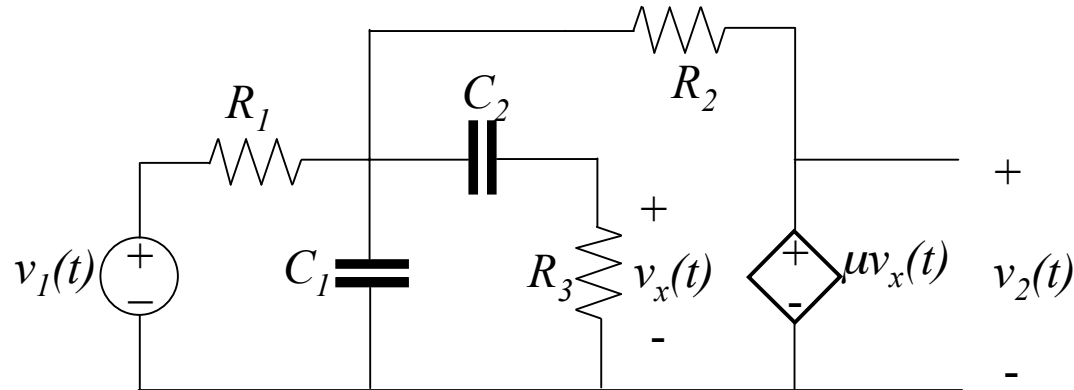
$$v_{zs}(t) = \left[3e^{-1000t} - 3e^{-3000t} \right] u(t)$$

$$V_{zi}(s) = \frac{1.5s}{(s+1000)(s+3000)} = \frac{-0.75}{s+1000} + \frac{2.25}{s+3000}$$

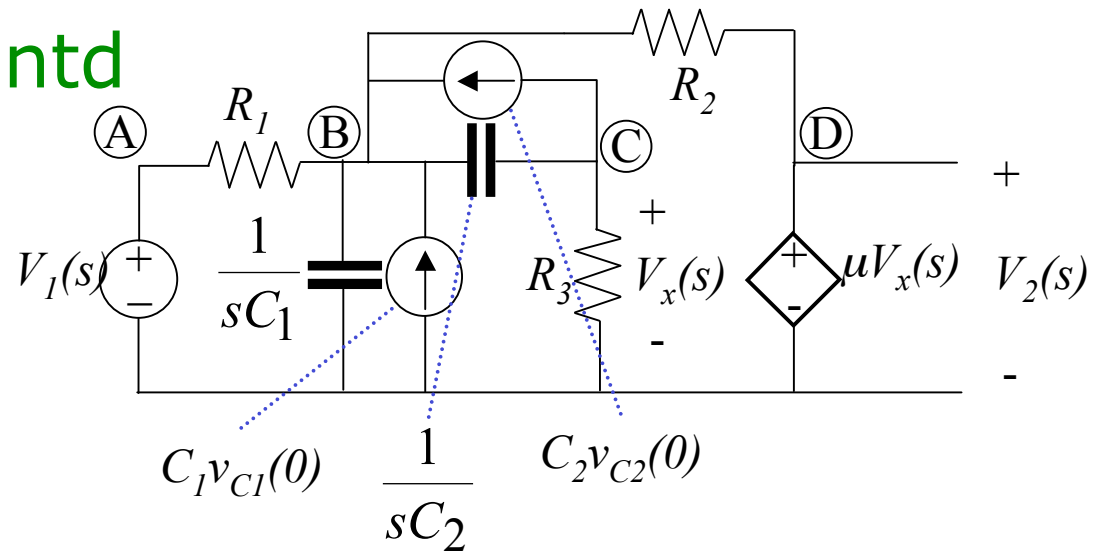
$$v_{zi}(t) = \left[-0.75e^{-1000t} + 2.25e^{-3000t} \right] u(t)$$

Example 10-11 T&R (not in 4th ed)

Formulate node voltage equations in s-domain



Example 10-11 contd



Node A: $V_A(s) = V_1(s)$

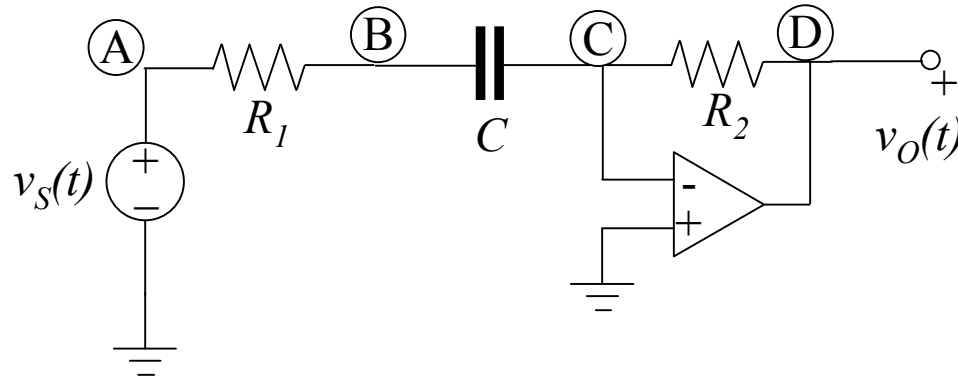
Node D: $V_D(s) = \mu V_x(s) = \mu V_C(s)$

Node B:
$$\frac{V_B(s) - V_A(s)}{R_1} + \frac{V_B(s) - V_D(s)}{R_2} + \frac{V_B(s)}{1/sC_1} + \frac{V_B(s) - V_C(s)}{1/sC_2} - C_1 v_{C1}(0) - C_2 v_{C2}(0) = 0$$

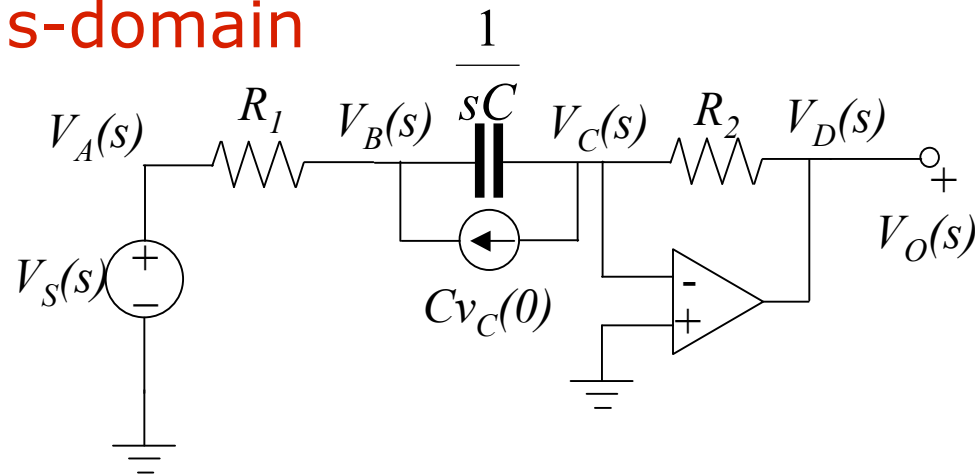
Node C:
$$-sC_2 V_B(s) + [sC_2 + G_3] V_C(s) = -C_2 v_{C2}(0)$$

Example 10-16 T&R (not in 4th ed)

Find $v_o(t)$ when $v_s(t)$ is a unit step $u(t)$ and $v_c(0)=0$



Convert to s-domain



Example 10-16 contd

Nodal Analysis

Node A: $V_A(s) = V_S(s)$

Node D: $V_D(s) = V_O(s)$

Node C: $V_C(s) = 0$

Node B: $(G_1 + sC)V_B(s) - G_1V_S(s) = Cv_C(0)$

Node C KCL: $-sCV_B(s) - G_2V_O(s) = -Cv_C(0)$

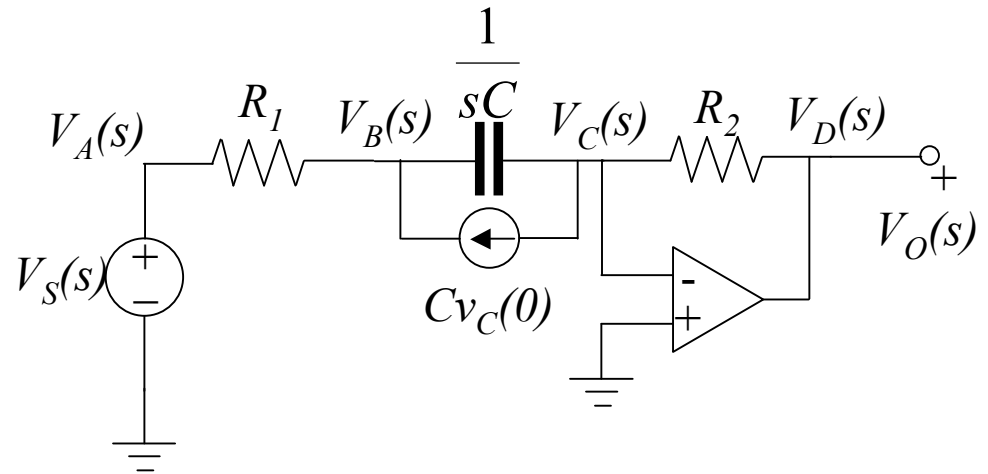
Solve for $V_O(s)$

$$V_O(s) = -\left[\frac{sG_1C}{G_1 + sC} \times \frac{1}{G_2}\right]V_S(s) = -\left[\frac{R_1}{R_2} \times \frac{s}{s + 1/R_1C}\right]V_S(s)$$

$$= -\left[\frac{R_1}{R_2} \times \frac{s}{s + 1/R_1C}\right] \frac{1}{s} = \frac{-R_1}{R_2} \times \frac{1}{s + 1/R_1C}$$

Invert LT

$$v_O(t) = \frac{-R_1}{R_2} e^{-\frac{t}{R_1C}} u(t)$$



Features of s-domain cct analysis

The response transform of a finite-dimensional, lumped-parameter linear cct with input being a sum of exponentials is a rational function and its inverse Laplace Transform is a sum of exponentials

The exponential modes are given by the poles of the response transform

Because the response is real, the poles are either real or occur in complex conjugate pairs

The natural modes are the zeros of the cct determinant and lead to the natural response

The forced poles are the poles of the input transform and lead to the forced response

Features of s-domain cct analysis

A cct is stable if all of its poles are located in the open left half of the complex s-plane

A key property of a system

Stability: the natural response dies away as $t \rightarrow \infty$

Bounded inputs yield bounded outputs

A cct composed of Rs, Cs and Ls will be at worst marginally stable

With Rs in the right place it will be stable

$Z(s)$ and $Y(s)$ both have no poles in $\text{Re}(s) > 0$

Impedances/admittances of RLC ccts are “Positive Real” or energy dissipating